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$$z = x^2y + 3xy^4$$

$$x = e^t$$

$$y = \sin t$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$= (2xy + 3y^4)e^t + (x^2 + 12xy^3) \cos t$$

$$= 2e^{2t} \sin t + 3e^t \sin^4 t + e^{2t} \cos t + 12e^t \sin^3 t \cos t$$

References